INTERNATIONAL MONETARY FUND

MYANMAR

February 11, 2020

STAFF REPORT FOR THE 2019 ARTICLE IV CONSULTATION—DEBT SUSTAINABILITY ANALYSIS

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Prepared by Staff of the International Monetary Fund and the International Development Association

Risk of external debt distress:	Low
Overall risk of debt distress:	Low
Granularity in the risk rating	Not applicable
Application of judgement:	No

Myanmar's risks of external debt and overall debt distress continue to be assessed as low. While the overall debt outlook remains positive—several vulnerabilities exist. These include a slowdown in FDI, exports, the aftermath from a natural disaster or from larger than expected contingent liabilities arising from the banking system or from a ramp up in PPPs. Increasing the export base, maximizing concessional loans and improving policy buffers would help keep the debt burden contained. Progress on the Rakhine state crisis has been slow and is affecting the outlook for external inflows. Limited monitoring of fiscal commitments and contingent liabilities poses significant fiscal risks. Infrastructure projects with high returns should be targeted keeping in mind that even a moderate increase in the deficit implies a recourse to central bank financing. Debt management capacity needs strengthening, and the authorities should remain vigilant about borrowing that leads to a rapid debt buildup. Strengthening the business environment and governance, including in the natural resource sector would raise the investment outlook and potential growth.

PUBLIC DEBT COVERAGE AND COUNTRY CLASSIFICATION

1. The coverage of public sector debt used in the DSA is consolidated public sector debt, government-guaranteed debt and social security funds. SOE debt is on lent and is therefore included in the coverage of public external debt.[1] There is no outstanding debt to the IMF.

Coverage of Public Sector Debt

1	Central government	X
2	State and local government	X
3	Other elements in the general government	
4	*Of which* : Social security fund	X
5	*Of which* : Extra budgetary funds (EBFs)	
6	Guarantees (to other entities in the public and private sector, including to SOEs)	X
7	Central bank (borrowed on behalf of the government)	
8	Non-guaranteed SOE debt	

Source: IMF staff estimates.

2. The LIC DSF determines the debt sustainability thresholds by calculating a composite indicator (CI). The CI, based on a weighted average of several factors such as the country's real GDP growth, remittances, international reserves, and world growth and the CPIA score.[2] The calculation of the CI is based on 10-year averages of the variables, across 5 years of historical data and 5 years of projection, and the corresponding CPIA. For Myanmar, the final debt carrying capacity classification for this DSA is medium. A summary of the thresholds used in the exercise are included in the table alongside.

Applicable Thresholds

APPLICABLE		APPLICABLE	
EXTERNAL debt burden thresholds		TOTAL public debt benchmark	
PV of debt, in percent of		PV of total public debt, in percent of GDP	55
Exports	180		
GDP	40		
Debt service, in percent of			
Exports	15		
Revenue	18		

Source: IMF staff estimates.

[1] The 2017 public debt management Law states that SOES can borrow directly from state-owned banks or benefit of on lending from the government. By law, SOE's cannot borrow directly from external creditors.

[2] The details on the methodology can be found in the new LIC-DSF guidance note: https://www.imf.org/en/Publications/Policy-Papers/Issues/2018/02/14/pp122617guidance-note-on-lic-dsf.

BACKGROUND ON DEBT

3. Myanmar's total public debt is estimated to be 38.1 percent of GDP as of FY2018/19.[3] Public debt in this DSA covers public domestic debt (61.8 percent of total public debt in FY2018/19) and public and publicly guaranteed (PPG) external debt (38.2 percent of total public debt in FY2018/19).[4] Domestic debt comprises T-bills and T-bonds, a large share of which—mostly 3-month T-bills—is held by the central bank. The largest share of PPG external debt is held by bilateral creditors amongst which China and Japan are the largest creditors. International Development Association (IDA) and the Asian Development Bank (ADB) are the largest multilateral creditors.

Total Public and Publicly Guaranteed Debt

(Estimated End-August 2019)	(USD millions)	In percent of total
Bilateral Creditors	**8.1**	**30.6**
Paris Club	4.2	16.0
Of which: Japan	2.9	10.9
Non-Paris Club	3.9	14.6
Of which: China	3.4	13.0
Multilateral Creditors	**2.0**	**7.5**
Of which: ADB	0.6	2.2
Of which: IDA	1.4	5.2
Commercial Creditors	**0.0**	**0.1**
Domestic Debt	**16.3**	**61.8**
T-Bills	11.7	44.2
T-Bonds	4.7	17.6
Total	**26.4**	**100.0**

Source: Myanmar authorities.

4. Total private external debt is estimated to be 13.1 percent of GDP as of FY2018/19. The estimated total private external debt stock has been revised to reflect the findings of private external debt benefitting from the intensive ongoing TA in external sector statistics Thus, total external debt, a total of PPG external and private external debt, is estimated to be 27.9 percent of GDP as of FY2018/19.

5. The authorities continue to seek external financing on concessional terms, however, reliance on CBM financing persists. New loans with bilateral and multilateral creditors have been signed on highly concessional terms for financing infrastructure and other projects ranging from power, electrification, and transportation to health. During the transition period (April 2018 – September 2018) and in the new fiscal year (October 2018 – September 2019) loans totaling approximately US$ 843.0 million have been signed. With revenues broadly in line with expectations, the fiscal deficit is estimated at about 3.5 percent of GDP in FY2018/19, compared to 3 percent in FY2017/18. The unexpected rapid pickup in the deficit toward the end of the year raised sharply CBM financing to 8.4 percent of last year's reserve money in FY2018/19, higher than the authorities' plan. However, the authorities are still committed to phasing out CBM financing through greater treasury securities issuances.

[3] Myanmar's fiscal year has changed from April-March, to October-September. Following a six-month transition period, from April 1, 2018 to September 30, 2018, the new fiscal year started October 2018. This first year of projection in this DSA is FY2018/19. FY2018/19 covers the period October 2018-September 2019.

[4] For Myanmar, the external debt definition used in this DSA is based on residency.

6. Contingent liabilities should be closely monitored. Staff estimate current off-balance sheet liabilities related to PPPs and PPAs are approximately 3.2 percent of GDP. Using the standard methodology under the IMF/WB LIC DSA framework translates this to a shock of 1.09 percent of GDP as part of the contingent liabilities.[5] Banks have continued to make progress converting from overdraft into term loans and appear to have met the July 2019 target of 30 percent of total lending set by the CBM. However, with capital positions and profitability low in most private banks, banks will need to recapitalize through time. Data limitations hinder a full assessment of potential NPLs and recapitalization needs, however, the uneven loan loss recognition and inadequate provisions, large exposures, and low capital position of banks indicate potential systemic concerns. The standard shock covering 5 percent of GDP has been added to the analysis to account for potential recapitalization needs.

Myanmar Off-Balance-Sheet Debt related to PPPs and PPAs

	Million of USD	Percent of 2018 GDP
Electricity	1874.7	2.8
of which electricity BOT	*1399.6*	*2.1*
of which electricity JV/BOT	*475.1*	*0.7*
Gas	131.8	0.2
Ports	110.0	0.2
Total	**2116.4**	**3.2**

Sources: Myanmar authorities; and PPI database.

Coverage of Contingent Liabilities

1 The country's coverage of public debt	The central, state, and local governments plus social security, government-guaranteed debt		
	Default	Used for the analysis	Reasons for deviations from the default settings
2 Other elements of the general government not captured in 1.	0 percent of GDP	0	
3 SoE's debt (guaranteed and not guaranteed by the government) 1/	2 percent of GDP	0	By law: SOEs cannot borrow directly externally
4 PPP	35 percent of PPP stock	1.09	
5 Financial market (the default value of 5 percent of GDP is the minimum value)	5 percent of GDP	5	
Total (2+3+4+5) (in percent of GDP)		6.1	

Source: IMF staff estimates.

MACRO FORECASTS IN THE BASELINE SCENARIO

7. The assumptions in the baseline scenario are consistent with the macroeconomic framework presented in the staff report. The main assumptions in the baseline are:

[5] A PPP shock in the LIC DSA framework is applicable only when the PPP capital stock is larger than 3 percent of GDP; per the latest data from the World Bank's PPP database, Myanmar PPP stock is estimated to be 1.42 percent of GDP.

- **Real GDP growth.** Economic activity remained below potential in 2018/19. Growth appears to have slowed further to 6.4 percent in FY2018/19 from 6.5 percent in the six-month transition budget, on the back of weaker investment and a lower than expected fiscal stimulus. Investment demand remains subdued due to moderating credit growth in the backdrop of a correction in real estate prices. Over the medium term, growth is expected to gradually rise as infrastructure investment accelerates. However, the pace would be slower than envisaged in the 2018 Article IV as bank restructuring begins in earnest post-election, slowing credit and GDP growth in line with the experience of countries that have experienced credit booms.

Key Macroeconomic Assumptions, FY 2019-FY 2024 1/
(Average)

	Baseline	Previous DSA
Real GDP growth (in percent)	6.2	6.7
Inflation (percent change, y/y)	7.0	6.5
Primary fiscal balance (in percent of GDP)	-2.1	-1.9
Overall fiscal balance (in percent of GDP)	-3.9	-3.6
Current account (in percent of GDP)	-4.7	-4.8
FDI (in percent of GDP)	2.9	4.1

Source: IMF staff estimates.
1/ FY2019 runs from October 2018 to September 2019. Thereafter, all fiscal years are on a October to September basis.

- **Inflation.** Headline inflation stood at 8.6 percent at end-September 2019 on higher electricity tariffs and food and fuel prices but is expected to moderate. Inflation is expected to fall to 6-7 percent range in the medium term as the one-off increase in electricity tariff and pressures from rising food prices abates.

- **Current account.** Subdued economic activity has narrowed the current account deficit. Imports contracted sharply on lower FDI related imports, decline in iron and steel imports due to a slowdown in construction and one-off factors related to changes in automobile import regulations. In contrast, exports held up, led by garment manufacturing and natural gas, despite global trade tensions, the slowdown in China and disruptions in the China-Myanmar trade corridor. The narrower deficit offset weaker FDI and other inflows. The current account deficit is projected to increase to 5 percent of GDP over the short to medium term reflecting a slowdown in gas exports and a relatively stronger demand for imports including oil.

- **External financing.** Project financing from large creditors is currently in place, however, budget financing from creditors remains uncertain. Multilateral financing is expected to remain stable, while bilateral financing is projected to pick up in the near term. While FDI inflows have continued to decline, FDI approvals, a leading indicator of inflows has seen a small uptick recently suggesting a slight recovery in inflows over the short to medium term (see Box 1). Little progress has been made in solving the humanitarian crisis and according to the UN, security has deteriorated in border areas. With elections due in 2020, in staff's view, prospects for progress are limited. The economic outlook is heavily dependent on external financing channels, and progress on the humanitarian crisis remains critical for donor financing.

- **Fiscal outlook.** Revenues are expected to remain stable as gains from tax compliance offsetting declining gas revenues, while the tax to GDP ratio remains well below peers. Fiscal spending will focus on social and capital spending which will help close the SDG gaps in education, health and infrastructure. Recent underspending on investment projects partly reflects implementation

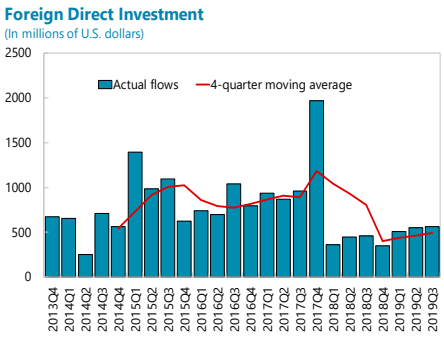
constraints and optimistic planning, while some acceleration is expected in the short term as many projects are entering the implementation phase. Fiscal deficit is projected to increase from previous years and remain around 4½ of GDP in the medium term, with primary deficit around 2½ percent of GDP.

- **Realism of the baseline.** The PPG external debt-to-GDP ratio follows a similar path compared to the previous DSA (published 2019) but is shifted downwards compared with the DSA from five years ago (published 2014) given the large debt relief. Cross country experience suggests that the baseline fiscal expansion is feasible, and the projected growth path is in line with a fiscal multiplier of 0.2.[6]

Box 1. Foreign Direct Investment to Myanmar (2015 – 18)[1/]

Myanmar's FDI statistics are compiled by the Directorate of Investment and Company Administration (DICA) under the Ministry of Investment and Foreign Economic Relations. DICA has extensively improved the FDI data collection and compilation process over the past three years and is currently producing FDI statistics in line with sixth edition of the Balance of Payments Manual. Funded by Japan, technical assistance in this area continues from the IMF's Statistics Department and Capacity Development Office in Thailand.

Total FDI in 2018 reached US$ 1.6 billion, compared with US$ 4.7 billion (2017) and US$ 3.9 billion (2016) respectively. This significant drop is attributed to several factors ranging from saturation in investments in key sectors that have received significant investments in the last 3 to 4 years, completion of large projects and the overall slowdown of foreign investments, particularly to emerging and frontier markets, in 2018. In addition, there is a notable reduction in reinvested earnings and a higher repayment of intercompany debt in 2018, particularly compared to 2017. Japan, China, Singapore and Thailand continue to remain the top sources of FDI in 2018.



Sources: Mvanmar authorities: and IMF staff calculations.

[6] Public/Private investment rate charts are not available in the current DSA from data limitations. Technical assistance from the IMF and various partners continue is ongoing to strengthen macroeconomic data (Appendix VII).

Box 1. Foreign Direct Investment to Myanmar (2015 – 18)[1] (concluded)

A detailed sectoral breakdown of FDI transactions for 2017 and 2018 (Table 1) highlights that except for banking, services (excluding banking), and airport and aviation sectors, there has been a notable reduction in FDI to all other major sectors in 2018. Key sectors, such as manufacturing, oil and gas, mining, telecommunications and hotels, received limited FDI in 2018 compared to the previous three years, indicating some saturation of investments. Significant intercompany debt has been repaid and reinvested earnings recorded a negative investment in some sectors. The mining and oil and gas sectors recorded disinvestments as well as significant intercompany debt repayments in 2018. Data also indicate that financial services, services sector (excluding banking and finance) and real estate & construction are some of the emerging sectors that have received notable investments in the past few years.

Table 1. FDI transactions
(US$ millions, calendar year basis)

	2015	2016	2017	2018
Equity – Paid up Capital	2,132	2,842	2,525	2,080
Equity - Reinvested	-6	-129	419	109
Intercompany debt	2,148	1,168	1,790	-580
Total FDI	**4,274**	**3,881**	**4,733**	**1,610**

Source: Myanmar authorities.

While FDI inflows have continued to decline, FDI approvals, a leading indicator of inflows has seen a small uptick recently suggesting a slight recovery in inflows over the short to medium term. In addition, FDI inflows are expected to improve related to the development of the A6 block which has estimated reserves of 2-3 trillion cubic feet of natural gas.[2/]

[1] Data (calendar year basis) are based on annual surveys and is supplemented with administrative data received from the Foreign Exchange Management department of the central bank, Thilawa Special Economic Zone, Myanmar Oil and Gas Enterprise and the Ministry of Electricity and Energy. Currently, the Foreign Direct Investment Survey (FDIS) covers over 3,500 active direct investment enterprises.

[2] Oil and Gas journal; September 24, 2018. Available via https://www.ogj.com/exploration-development/discoveries/article/17296507/total-updates-shwe-yee-htun2-gas-appraisal-offshore-myanmar.

EXTERNAL AND PUBLIC DEBT SUSTAINABILITY

8. All external PPG debt indicators remain below the policy relevant thresholds in the baseline scenario (Figure 1). The PV of external debt-to GDP ratio is gradually declining from 12.0 percent of GDP in FY2018/19 to around 10.0 percent of GDP over the projection period. Standardized stress tests show that slowdown in exports and FDI negatively impact the PV of debt-to-GDP ratio, PV of debt-to-exports ratio and the debt service-to-exports ratio (Figure 2) highlighting the need to expand the export base, the importance of the new bidding round and

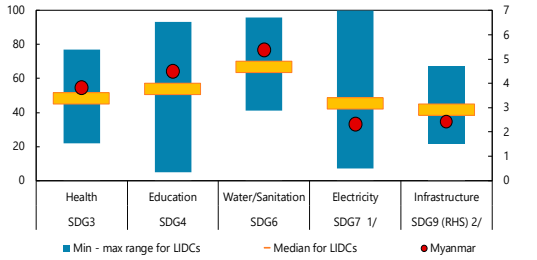
taking the right steps to improve the business environment. The debt service-to-revenue ratio is affected by depreciation. The authorities need to build up policy buffers, particularly domestic revenues and foreign reserves, and continue their efforts with structural reforms to improve growth potential and resilience and promote economic diversification.

9. The PV of total public debt-to-GDP ratio lies below the public debt benchmark (Figure 2). The standardized sensitivity analysis shows that the largest shock that pushes the PV of public debt-to-GDP in FY2027/28 to reach 55.0 percent of GDP is the natural disaster shock (Figure 2, Table 4).[7] Myanmar is exposed to a range of natural disasters accompanied by high economic and social costs, with significant impact on the poor and most vulnerable.[8] Annual expected losses from natural hazards are one of the highest among all countries in Southeast Asia (World Bank, GFDRR). To effectively mitigate the impact of climate-related disasters, the authorities need to enhance preparedness and response ability through addressing weaknesses in ex-ante resilience and ex-post adaptive capacity. The vulnerability to this shock also highlights the need for stronger growth in the medium term and the importance of building fiscal and external buffers.

STAFF SCENARIO ANALYSIS[9]

10. The planned scaling up of infrastructure projects is key to accelerating growth but entails significant fiscal costs and risks. Myanmar is planning to undertake large investment projects in the near to medium term to address sizeable infrastructure gaps in electricity, gas, roads and railways, and ports as part of their commitment to reaching SDGs as detailed in the Myanmar Sustainable Development Plan. Scaling up public investment requires additional financing, which would have important implications for public debt dynamics. Some of these projects are also being proposed as Public-Private Partnerships (PPPs), which may bring efficiency gains, but can also entail substantial fiscal risks where they involve contingent liabilities, including from sovereign guarantees. Staff assess the macro-fiscal impact of these

Myanmar: Selected Sustainable Development Goals (SDGs) Performance



Source: IMF staff calculations using Sachs, J., Schmidt-Traub, G., Kroll, C., Durand-Delacre, D. and Teksoz, K. (2017): SDG Index and Dashboards Report 2018. New York: Bertelsmann Stiftung and Sustainable Development Solutions Network (SDSN).
1/ Indicator: Access to electricity (percentage of population).
2/ Indicator: Quality of Infrastructure.

[7] One-off shock of 10 percentage points of GDP to debt-GDP ratio in the second year of the projection period (2019 for this case). Real GDP growth and exports are lowered by 1.5 and 3.5 percentage points, respectively, in the year of the shock.

[8] Risks from natural disasters are ever present in Myanmar incurring large economic losses in the aftermath. Historically, these losses have been estimated to be around 2 percent of GDP annually (between 2006–15) and were also highlighted, in the previous DSA, as the largest shocks that affect debt sustainability (IMF/WB DSA 2018).

[9] The staff scenario analysis is an illustrative exercise. It differs from the standard shock analyses in the LIC DSA framework where shocks are standardized across countries to facilitate cross country comparison.

infrastructure investments through a scenario analysis by considering the growth benefits of a ramp up of such investment resulting in increased public debt and fiscal risks.

11. The scaling up scenario is drawn from development partner studies on electricity and transportation infrastructure gaps and the authorities' intentions as signaled in the *Project Bank*. [10] Sectoral investment assumptions and their implications for the fiscal spending, investment flows, and GDP growth are as follows.

- **Electricity sector.** A recent World Bank study estimates that Myanmar needs to double the historical level of investment spending (both public and private) to meet its rapidly growing electricity demand.[11] In line with this study and projected needs, this scenario assumes that, for the public sector, (i) annual power purchasing agreement (PPA) payments to independent power producers (IPPs) will increase by US$800 million for an additional capacity of 1000 MW, enough to satisfy the nation's demand for electricity in the next 5 years, and (2) public sector capital spending will increase by US$500 million (doubling the historical level) for scaling up infrastructure in generation, transmission and distribution. Accordingly, private sector investments in generation by IPPs is expected to also double from historical levels which would provide the required additional generation capacity,

- **Transportation sector.** Myanmar's transportation infrastructure lags regional peers. In a 2016 study, Asian Development Bank estimates that the country needs to increase investments in the sector to 3-4 percent of GDP (up from 1-1.5 percent) in line with other countries at similar stages of development (e.g., China, Thailand, and Vietnam). [12] In this scenario, an additional spending of 2 percent of GDP each year is assumed. Furthermore, it is assumed that the additional spending is equally shared between direct budget spending and private sector (through PPPs).

- **Macroeconomic linkage.** Scaling up infrastructure investments will boost Myanmar's growth potential. WEO (2014) suggests that the contemporaneous effect of a 1 percentage point of GDP increase in public investment is a 0.25 percent increase in output, which gradually increases to about 0.5 percent four years after the shock.[13] As a result, under the assumed investment scaling up, GDP growth could be raised by 1.5–2.0 percent in the medium to long term.[14]

12. Under the scaling up scenario, the PV of public-debt ratio rises but remains under its indicative thresholds. In this scenario, the PV-of-public-debt to GDP ratio deteriorates significantly

[10] The scenario analysis focuses on the electricity and transportation sectors in Myanmar which directly link to SDGs and capture the main areas that significant public investment is envisaged. Data availability in other sectors is sparse.

[11] World Bank (2019). Myanmar: Energy Infrastructure Sector Assessment. *Unpublished*.

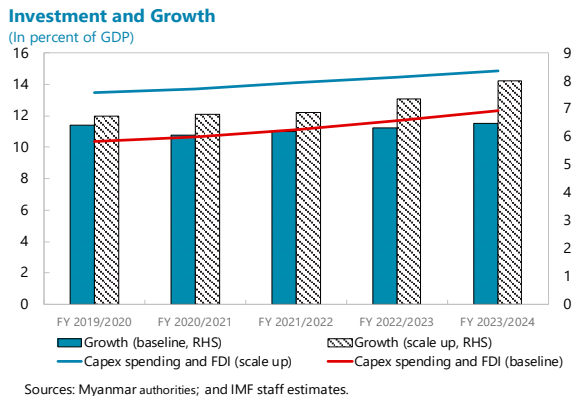
[12] Asian Development Bank (2016). Myanmar Transport Sector Policy Note: Summary for Decision Makers.

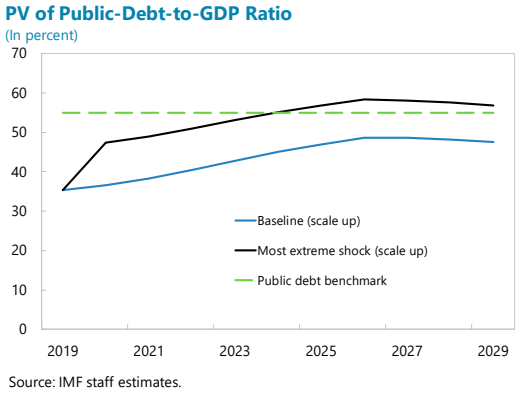
[13] This is broadly in line with the calibrated heterogeneous agent dynamic stochastic equilibrium model tailored to Myanmar. Macroeconomic and Distributional Implications of Financial Reforms in Myanmar, in Myanmar Selected Issues Paper, IMF Country Report 17/31.

[14] Prior estimates have suggested that Myanmar can achieve long-term growth in the 7-8 percent range. More details are available in Box 2 of Myanmar: 2016 Article IV Consultation (Country Report No. 17/30).

despite the assumption that the planned scaled up investments could raise growth to about 8 percent in the medium to long term. It could also push the PV-of-public-debt to GDP ratio, under the most extreme shock, to breach its indicative threshold in the medium term suggesting a worsening of the risk rating under the IMF/WB's LIC DSA framework.

Scaling Up Scenario





Sources: Myanmar authorities; and IMF staff estimates.

13. Although privately-financed investments help to address infrastructure bottlenecks, they are likely to create large fiscal commitments and in turn significant fiscal risks. The assumed additional PPP projects and PPA payments could result in long term fiscal commitments and explicit and implicit contingent liabilities. The PPA payment commitments assumed in this scenario not only have direct budget costs, but also create additional fiscal risks as payments for electricity purchased are measured in foreign currency and the government will assume the risk of exchange rate fluctuations. The limited ability of the government to track these contingent liabilities is particularly important given the need for foreign currency requirements and the level of reserves.

14. The higher public debt trajectory and fiscal risks resulting from scaling up highlights the need to strengthen public investment selection and management framework. To reap the maximum benefits on future growth from today's infrastructure investments, the authorities need to ensure that projects with the highest economic returns are selected and implemented. The first public investment program in Myanmar, the Project Bank, was launched in December 2019 with support from various development partners including the World Bank. Although still at its infancy, the Project Bank aims to eventually become the hub to identify and screen infrastructure projects and account for the fiscal liabilities from PPPs. To makes this happen, the authorities need to implement a public investment management framework that ensures value for money assessments and sound fiscal risk management. The authorities are also committed to building the capacity of the Ministry of Planning and Finance to analyze and select these infrastructure projects.

Box 2. Energy Infrastructure Sector Investment

Myanmar has the lowest electrification rate in Southeast Asia with an access rate of 50 percent nationwide. Electricity consumption in Myanmar is only about one-tenth of the world's average. Access to modern fuels for daily activities is limited to mostly urban areas, where only a third of the population lives. It is expected that demand for electricity will only increase given the favorable medium-term outlook. The World Bank estimates that consumption will grow at an average annual rate of 11.1 percent until 2030 reaching 12.6 GW by 2030—a significant increase from the current level of 3.6 GW. In addition, the outlook for domestic gas production is on the decline starting 2020/21 -although gas production from the A6 block could help reduce the decline in the medium term. To achieve its energy goals, the authorities are aiming to attract private investments and increase public investments in power and gas sectors. Investments in these sectors are estimated to be around US$2 billion per year for electricity and about US$10 billion for gas over the medium term.

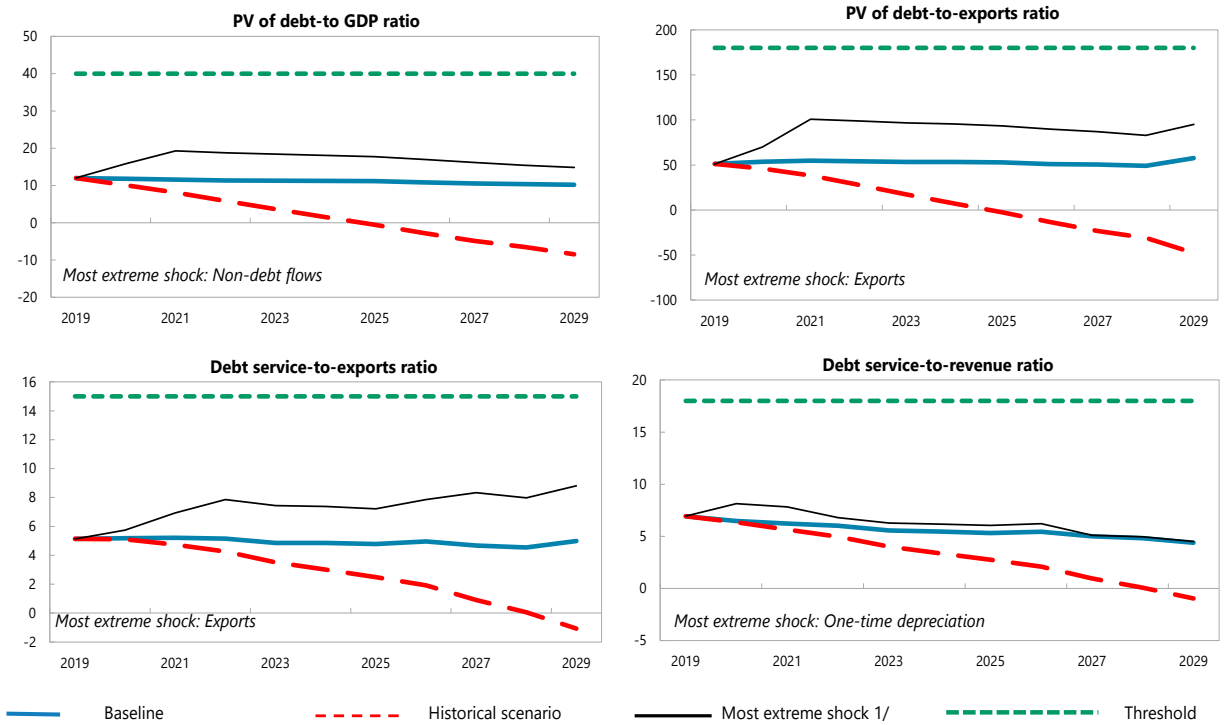
ASSESSMENT OF RISK RATING

15. Myanmar's risk of external debt distress is low as is the overall risk of debt distress. While the overall debt outlook remains positive, it remains vulnerable to slowdown in FDI, exports, and natural disasters. Progress on the Rakhine state crisis has been slow and is affecting the outlook for investment inflows and donor financing. More immediate concerns include larger than expected contingent liabilities arising from the banking system or from a rapid ramp up in PPPs. The authorities should remain vigilant of the potential impact from the fragilities in the banking system and put in place frameworks that better monitor fiscal risks incurred from PPPs. Moreover, currently, contingent liabilities are concentrated in a few sectors and thus establishing a monitoring framework at this stage is opportune. In this context, strengthening debt management capacity remains priority. To avoid a recourse to central bank financing, the authorities should embark on a medium-term revenue strategy underpinned by a revenue target and comprehensive tax policy reforms building on the administrative reforms. With low level of reserves, in year demand for FX requirements to service debt could put further depreciation pressures on the kyat. Thus, supporting the development of formal FX markets by addressing regulatory gaps, encouraging the availability of hedging instruments, and improving the availability of timely market information is key. Over the medium term, strengthening the business environment and improving governance would raise the investment outlook and potential growth. Over the longer term, the need to strengthen buffers against natural disasters by increasing fiscal and external buffers is emphasized which in turn will also strengthen debt sustainability.

Authorities' Views

16. The authorities broadly agreed with staff's assessment of the debt sustainability analysis. In the context of the proposed infrastructure scaling up, the authorities appreciated the staff analysis and presentation on risks from scaling up. They reiterated the commitment to phasing out CBM financing of the budget and explained the overshooting of the target at end FY 2018/19 as temporary. Although the "Project Bank" has just been launched, they intended to make it instrumental in managing public investment including PPPs. They also welcome oncoming Bank and Fund TA to support them in strengthening SEE governance and the management of fiscal risks.

Figure 1. Myanmar: Indicators of Public and Publicly Guaranteed External Debt Under Alternatives Scenarios, 2019–2029 1/ 2/



Customization of Default Settings	Size	Interactions
Tailored Stress		
Combined CL	Yes	
Natural disaster	No	No
Commodity price	n.a.	n.a.
Market financing	n.a.	n.a.

Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

Borrowing assumptions on additional financing needs resulting from the stress tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	2.6%	5.0%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	23	20
Avg. grace period	6	5

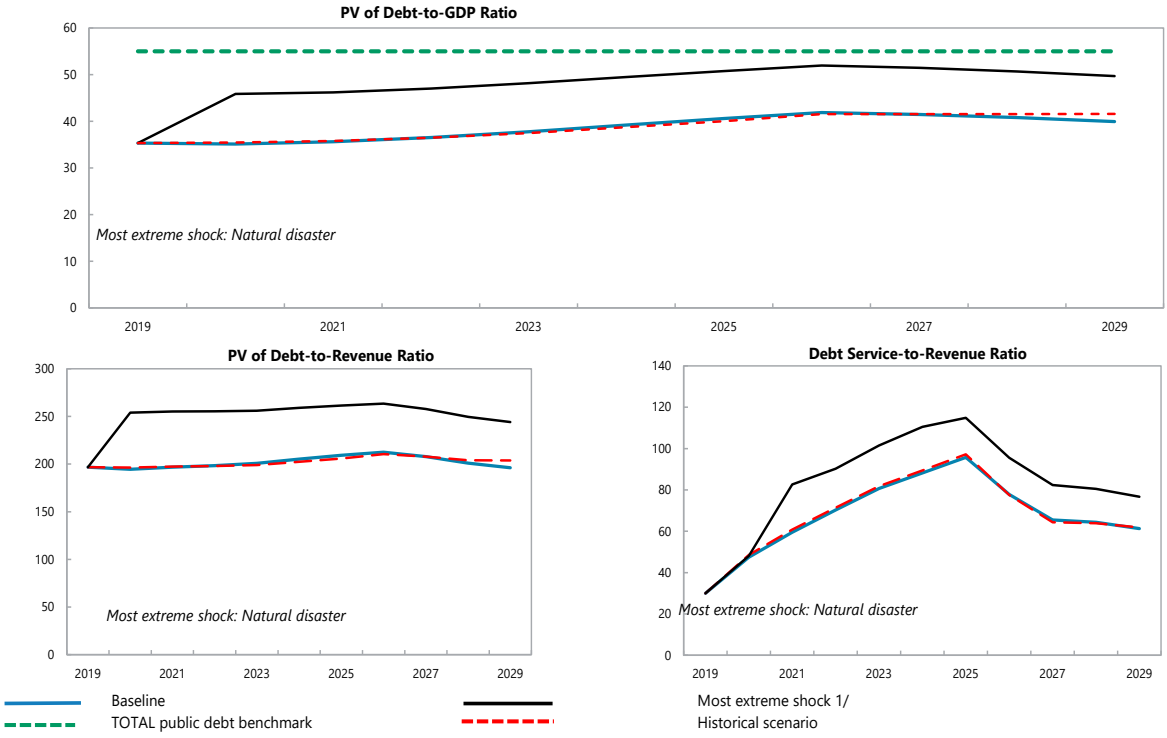
* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2029. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

2/ The magnitude of shocks used for the commodity price shock stress test are based on the commodity prices outlook prepared by the IMF research department.

Figure 2. Myanmar: Indicators of Public Debt Under Alternative Scenarios, 2019–29 1/



Borrowing assumptions on additional financing needs resulting from the stress tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	12%	12%
Domestic medium and long-term	52%	52%
Domestic short-term	36%	36%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	2.6%	5.0%
Avg. maturity (incl. grace period)	23	20
Avg. grace period	6	5
Domestic MLT debt		
Avg. real interest rate on new borrowing	0.4%	3.0%
Avg. maturity (incl. grace period)	4	4
Avg. grace period	1	1
Domestic short-term debt		
Avg. real interest rate	0.0%	0.0%

* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

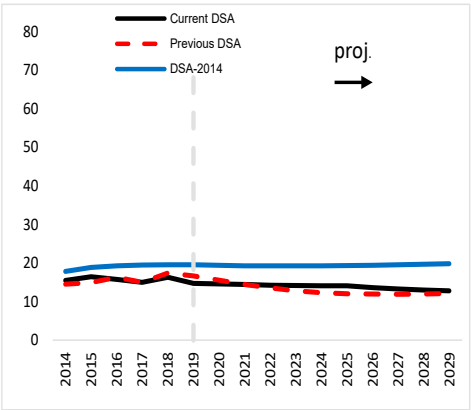
Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2029. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

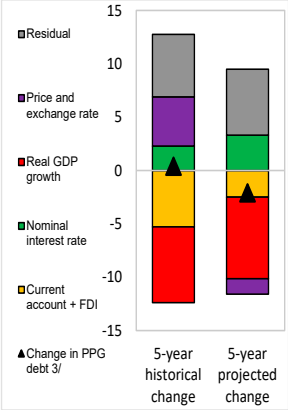
Figure 3. Myanmar: Drivers of Debt Dynamics—Baseline Scenario

External Debt

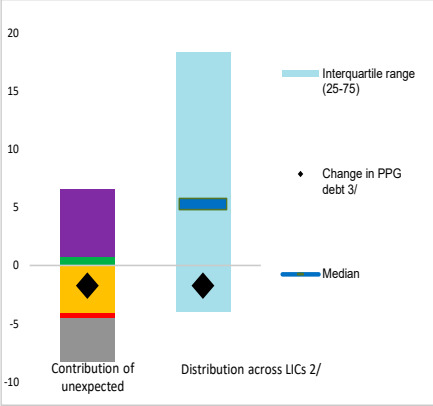
Gross Nominal PPG External Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)

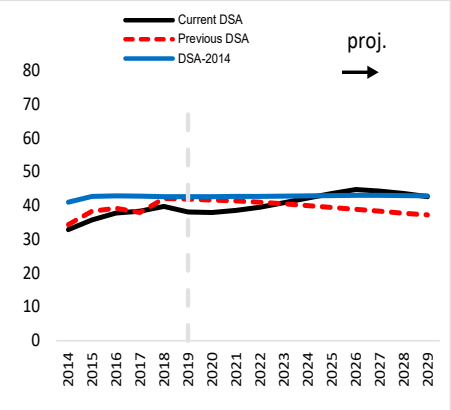


Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)

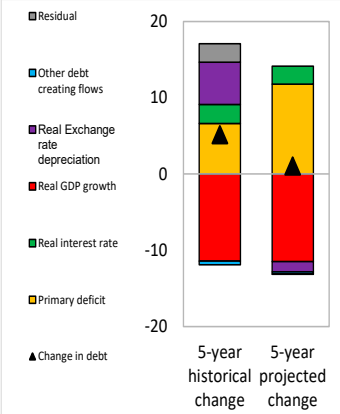


Public Debt

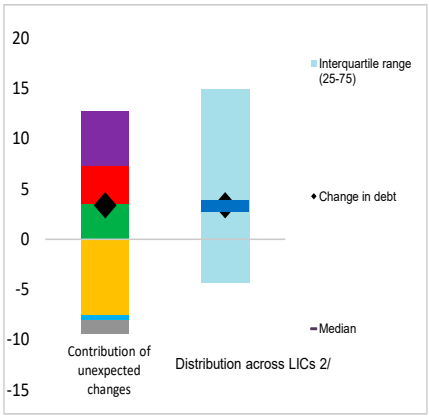
Gross Nominal Public Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)



Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)



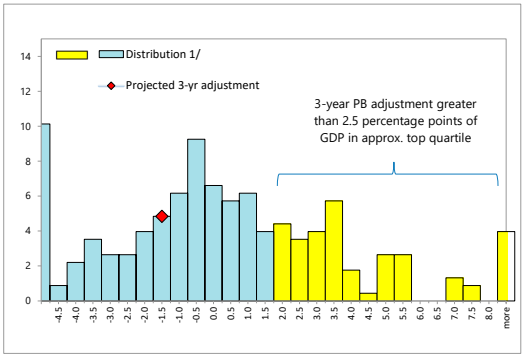
1/ Difference between anticipated and actual contributions on debt ratios.

2/ Distribution across LICs for which LIC DSAs were produced.

3/ Given the relatively low private external debt for average low-income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.

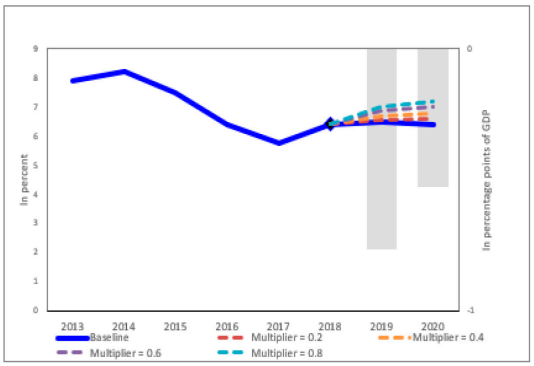
Figure 4. Myanmar: Realism Tools

3-Year Adjustment in Primary Balance (Percentage points of GDP)



1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

Fiscal Adjustment and Possible Growth Paths 1/



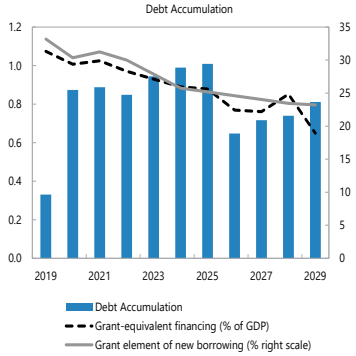
1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

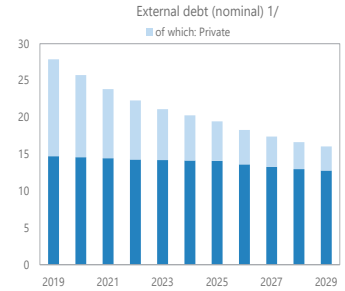
Table 1. Myanmar: External Debt Sustainability Framework, Baseline Scenario, 2016–39

(In percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 8/	
	2016	2017	2018	2019	2020	2021	2022	2023	2024	2029	2039	Historical	Projections
External debt (nominal) 1/	**20.9**	**28.9**	**30.9**	**27.9**	**25.7**	**23.8**	**22.3**	**21.1**	**20.3**	**16.1**	**15.7**	**17.9**	**20.8**
of which: public and publicly guaranteed (PPG)	**15.7**	**15.0**	**16.3**	**14.7**	**14.6**	**14.5**	**14.3**	**14.2**	**14.1**	**12.8**	**14.5**	**13.5**	**13.9**
Change in external debt	-0.5	8.0	2.0	-3.1	-2.1	-1.9	-1.5	-1.2	-0.8	-0.6	0.2		
Identified net debt-creating flows	**0.0**	**0.3**	**-3.1**	**-2.8**	**-1.5**	**-1.1**	**-0.8**	**-0.6**	**-0.5**	**0.2**	**0.6**	**-2.7**	**-0.8**
Non-interest current account deficit	**3.8**	**6.0**	**3.6**	**1.2**	**2.5**	**2.8**	**3.4**	**3.8**	**4.2**	**4.9**	**4.6**	**2.0**	**3.7**
Deficit in balance of goods and services	4.4	7.5	5.1	3.0	3.7	4.1	4.5	5.0	5.2	6.0	5.7	**1.0**	**4.9**
Exports	21.3	21.5	23.4	23.5	22.0	21.2	21.1	21.1	21.1	17.7	22.3		
Imports	25.8	29.0	28.5	26.5	25.7	25.3	25.5	26.1	26.3	23.6	28.0		
Net current transfers (negative = inflow)	-3.4	-3.7	-3.8	-3.8	-3.2	-3.1	-3.0	-2.9	-2.8	-2.3	-1.7	**-2.0**	**-2.8**
of which: official	-0.5	-0.6	-0.7	-0.6	-0.4	-0.4	-0.4	-0.4	-0.4	-0.3	-0.3		
Other current account flows (negative = net inflow)	2.7	2.2	2.3	2.0	2.0	1.9	1.9	1.7	1.7	1.2	0.6	**3.0**	**1.7**
Net FDI (negative = inflow)	**-5.1**	**-5.8**	**-4.8**	**-2.9**	**-3.1**	**-3.2**	**-3.5**	**-3.6**	**-4.0**	**-4.2**	**-3.6**	**-4.1**	**-3.7**
Endogenous debt dynamics 2/	**1.4**	**0.1**	**-1.8**	**-1.1**	**-0.9**	**-0.8**	**-0.8**	**-0.8**	**-0.7**	**-0.6**	**-0.4**		
Contribution from nominal interest rate	0.4	0.5	0.6	0.8	0.7	0.6	0.6	0.5	0.5	0.4	0.4		
Contribution from real GDP growth	-1.4	-1.2	-1.7	-2.0	-1.6	-1.4	-1.3	-1.3	-1.3	-1.0	-0.8		
Contribution from price and exchange rate changes	2.4	0.8	-0.7	...	...	...	...	...	...	...	...		
Residual 3/	**-0.5**	**7.7**	**5.1**	**-0.2**	**-0.7**	**-0.8**	**-0.7**	**-0.6**	**-0.3**	**-0.8**	**-0.4**	**4.0**	**-0.6**
of which: exceptional financing	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Sustainability indicators													
PV of PPG external debt-to-GDP ratio	**...**	**...**	**11.9**	**12.0**	**11.8**	**11.6**	**11.4**	**11.3**	**11.2**	**10.2**	**11.7**		
PV of PPG external debt-to-exports ratio	**...**	**...**	**51.0**	**51.2**	**53.6**	**54.7**	**54.0**	**53.4**	**53.3**	**57.7**	**52.5**		
PPG debt service-to-exports ratio	**4.4**	**4.7**	**5.1**	**5.1**	**5.2**	**5.2**	**5.2**	**4.9**	**4.9**	**5.0**	**4.1**		
PPG debt service-to-revenue ratio	**4.9**	**5.6**	**6.4**	**6.9**	**6.5**	**6.2**	**6.0**	**5.6**	**5.5**	**4.4**	**4.1**		
Gross external financing need (Million of U.S. dollars)	2812.7	4428.2	3403.4	3429.2	4271.7	4534.1	4536.6	4649.0	4637.5	5862.3	10523.3		
Key macroeconomic assumptions													
Real GDP growth (in percent)	6.4	5.8	6.4	6.5	6.4	6.0	6.2	6.3	6.5	6.5	5.8	**6.4**	**6.4**
GDP deflator in US dollar terms (change in percent)	-10.2	-3.7	2.6	-4.2	2.6	3.2	3.0	2.7	2.7	2.6	2.3	**4.3**	**2.1**
Effective interest rate (percent) 4/	1.7	2.3	2.3	2.8	2.8	2.7	2.7	2.6	2.6	2.7	2.7	**4.7**	**2.7**
Growth of exports of G&S (US dollar terms, in percent)	-10.8	2.5	18.8	2.4	2.4	5.4	8.7	9.6	8.9	-8.2	18.5	**8.9**	**6.0**
Growth of imports of G&S (US dollar terms, in percent)	-0.5	14.7	7.2	-5.2	6.0	7.6	10.5	11.6	10.2	-2.4	13.2	**15.4**	**6.9**
Grant element of new public sector borrowing (in percent)	...	...	...	33.2	30.4	31.2	30.0	27.9	25.8	23.2	22.4	**...**	**27.2**
Government revenues (excluding grants, in percent of GDP)	19.1	17.9	18.5	17.4	17.7	17.7	18.0	18.4	18.7	20.1	22.3	**16.3**	**18.7**
Aid flows (in Million of US dollars) 5/	647.8	658.6	1223.0	985.6	1076.3	1271.2	1207.9	1190.7	1168.9	1187.9	3004.2		
Grant-equivalent financing (in percent of GDP) 6/	...	...	...	1.1	1.0	1.0	1.0	0.9	0.9	0.6	0.7	**...**	**0.9**
Grant-equivalent financing (in percent of external financing) 6/	...	...	...	53.2	42.0	42.7	42.2	40.4	38.7	36.0	32.3	**...**	**41.5**
Nominal GDP (Million of US dollars)	60,411	61,504	67,145	68,538	74,802	81,868	89,552	97,789	106,919	166,772	386,548		
Nominal dollar GDP growth	-4.4	1.8	9.2	2.1	9.1	9.4	9.4	9.2	9.3	9.2	8.3	**10.8**	**8.6**
Memorandum items:													
PV of external debt 7/	...	...	26.5	25.1	22.9	21.0	19.4	18.2	17.3	13.5	12.9		
In percent of exports	...	...	113.5	107.2	104.2	98.9	92.1	86.0	82.4	76.1	58.0		
Total external debt service-to-exports ratio	27.7	32.4	27.0	28.6	28.5	27.8	24.4	21.6	19.5	15.5	7.5		
PV of PPG external debt (in Million of US dollars)			8003.9	8225.8	8824.8	9488.7	10182.9	11028.6	11996.5	17021.3	45275.9		
(PVt-PVt-1)/GDPt-1 (in percent)				0.3	0.9	0.9	0.8	0.9	1.0	0.8	1.3		
Non-interest current account deficit that stabilizes debt ratio	4.3	-2.0	1.6	4.2	4.6	4.8	4.9	5.0	5.0	5.5	4.4		

Definition of external/domestic debt	Residency-based
Is there a material difference between the two criteria?	No





Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g) + εα (1+r)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, ρ = growth rate of GDP deflator in U.S. dollar terms, ε=nominal appreciation of the local currency, and α= share of local currency-denominated external debt in total external debt.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

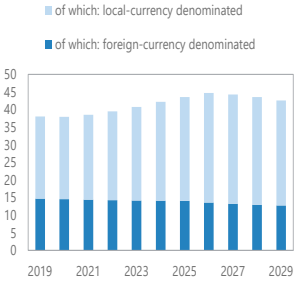
Table 2. Myanmar: Public Sector Debt Sustainability Framework, Baseline Scenario, 2016–39

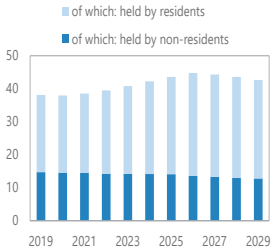
(In percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 6/	
	2016	2017	2018	2019	2020	2021	2022	2023	2024	2029	2039	Historical	Projections
Public sector debt 1/	**37.8**	**38.3**	**39.8**	**38.1**	**37.9**	**38.5**	**39.5**	**40.8**	**42.2**	**42.6**	**30.8**	**33.8**	**41.4**
of which: external debt	**15.7**	**15.0**	**16.3**	**14.7**	**14.6**	**14.5**	**14.3**	**14.2**	**14.1**	**12.8**	**14.5**	**13.5**	**13.9**
Change in public sector debt	2.0	0.5	1.4	-1.7	-0.1	0.6	1.0	1.3	1.4	-1.0	-1.0		
Identified debt-creating flows	**0.7**	**0.3**	**0.6**	**-0.9**	**0.0**	**0.8**	**1.1**	**1.4**	**1.5**	**-0.9**	**-1.0**	**-0.7**	**0.4**
Primary deficit	**2.6**	**1.4**	**1.1**	**1.8**	**2.4**	**2.6**	**2.5**	**2.5**	**2.5**	**1.3**	**0.4**	**1.9**	**2.1**
Revenue and grants	19.6	18.3	18.8	18.0	18.1	18.1	18.4	18.8	19.1	20.4	22.6	**16.5**	**19.1**
of which: grants	0.5	0.4	0.3	0.6	0.4	0.4	0.4	0.4	0.4	0.3	0.3		
Primary (noninterest) expenditure	22.1	19.6	19.9	19.8	20.4	20.7	20.9	21.3	21.6	21.6	23.0	**18.4**	**21.2**
Automatic debt dynamics	**-1.8**	**-0.9**	**-0.4**	**-2.6**	**-2.3**	**-1.8**	**-1.3**	**-1.0**	**-0.9**	**-2.1**	**-1.4**		
Contribution from interest rate/growth differential	-1.1	-1.9	-1.9	-2.6	-2.3	-1.8	-1.3	-1.0	-0.9	-2.1	-1.4		
of which: contribution from average real interest rate	1.0	0.2	0.4	-0.2	0.0	0.4	0.9	1.3	1.6	0.5	0.4		
of which: contribution from real GDP growth	-2.2	-2.1	-2.3	-2.4	-2.3	-2.2	-2.2	-2.3	-2.5	-2.7	-1.8		
Contribution from real exchange rate depreciation	-0.6	1.0	1.5	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**-0.1**	**-0.2**	**-0.1**	**-0.1**	**-0.1**	**-0.1**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**-0.4**	**0.0**
Privatization receipts (negative)	-0.1	-0.2	-0.1	-0.1	-0.1	-0.1	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (please specify)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Residual	**1.3**	**0.3**	**0.8**	**-0.8**	**-0.1**	**-0.2**	**-0.2**	**-0.1**	**-0.1**	**-0.1**	**-0.1**	**1.4**	**-0.2**
Sustainability indicators													
PV of public debt-to-GDP ratio 2/	**...**	**...**	**36.8**	**35.4**	**35.1**	**35.6**	**36.5**	**37.8**	**39.2**	**39.9**	**27.9**		
PV of public debt-to-revenue and grants ratio	**...**	**...**	**195.6**	**196.8**	**194.5**	**196.7**	**198.4**	**200.9**	**205.2**	**196.1**	**123.3**		
Debt service-to-revenue and grants ratio 3/	**28.2**	**29.4**	**28.8**	**30.0**	**47.4**	**59.5**	**70.3**	**80.7**	**88.2**	**61.3**	**29.5**		
Gross financing need 4/	8.0	6.5	6.4	7.2	10.9	13.3	15.4	17.6	19.3	13.7	7.1		
Key macroeconomic and fiscal assumptions													
Real GDP growth (in percent)	6.4	5.8	6.4	6.5	6.4	6.0	6.2	6.3	6.5	6.5	5.8	**6.4**	**6.4**
Average nominal interest rate on external debt (in percent)	2.6	2.3	3.7	2.0	2.0	2.0	2.1	2.1	2.1	2.5	2.7	**4.9**	**2.2**
Average real interest rate on domestic debt (in percent)	4.3	0.1	0.9	-1.0	-0.1	1.8	4.0	5.5	6.2	1.7	1.8	**-2.4**	**3.1**
Real exchange rate depreciation (in percent, + indicates depreciation)	-3.9	6.5	10.4	...	...	...	...	...	...	...	...	**-0.4**	**...**
Inflation rate (GDP deflator, in percent)	1.1	5.4	5.4	6.3	8.2	7.7	7.1	6.5	6.5	6.4	6.1	**5.5**	**6.8**
Growth of real primary spending (deflated by GDP deflator, in percent)	2.3	-6.2	7.9	6.0	9.8	7.6	7.2	8.0	8.2	5.4	6.4	**12.4**	**7.2**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	0.6	0.8	-0.4	3.5	2.5	2.0	1.5	1.2	1.1	2.2	1.4	**0.3**	**1.9**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	**Residency-based**
Is there a material difference between the two criteria?	**No**

Public sector debt 1/





Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The central, state, and local governments plus social security, government-guaranteed debt . Definition of external debt is Residency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Table 3. Myanmar: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2019–29

(In percent)

	Projections 1/										
	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029
PV of debt-to GDP ratio											
Baseline	12	12	12	11	11	11	11	11	11	10	10
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	12	10	8	6	4	2	-1	-3	-5	-7	-8
B. Bound Tests											
B1. Real GDP growth	12	12	12	12	12	12	12	11	11	11	11
B2. Primary balance	12	12	12	12	12	12	12	11	11	11	11
B3. Exports	12	14	18	17	17	17	17	16	15	15	14
B4. Other flows 3/	12	16	19	19	18	18	18	17	16	15	15
B5. Depreciation	12	15	12	11	11	11	11	11	11	11	11
B6. Combination of B1-B5	12	16	18	17	17	17	17	16	15	15	14
C. Tailored Tests											
C1. Combined contingent liabilities	12	13	13	13	13	13	13	12	12	12	12
C2. Natural disaster	12	13	13	14	14	14	14	14	14	14	14
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	40	40	40	40	40	40	40	40	40	40	40
PV of debt-to-exports ratio											
Baseline	51	54	55	54	53	53	53	51	50	49	58
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	51	46	38	28	17	7	-3	-13	-23	-31	-48
B. Bound Tests											
B1. Real GDP growth	51	54	55	54	53	53	53	51	50	49	58
B2. Primary balance	51	54	56	56	56	56	56	54	54	52	62
B3. Exports	51	70	101	99	97	96	93	90	87	83	95
B4. Other flows 3/	51	72	91	89	87	86	84	80	77	73	84
B5. Depreciation	51	54	43	43	43	43	43	42	42	41	49
B6. Combination of B1-B5	51	75	81	87	86	84	83	79	77	73	84
C. Tailored Tests											
C1. Combined contingent liabilities	51	57	59	59	60	60	60	59	59	57	68
C2. Natural disaster	51	61	65	66	66	68	68	68	68	67	80
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	180	180	180	180	180	180	180	180	180	180	180
Debt service-to-exports ratio											
Baseline	5	5	5	5	5	5	5	5	5	5	5
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	5	5	5	4	4	3	2	2	1	0	-1
B. Bound Tests											
B1. Real GDP growth	5	5	5	5	5	5	5	5	5	5	5
B2. Primary balance	5	5	5	5	5	5	5	5	5	5	5
B3. Exports	5	6	7	8	7	7	7	8	8	8	9
B4. Other flows 3/	5	5	6	7	6	6	6	7	7	7	8
B5. Depreciation	5	5	5	5	4	4	4	5	4	4	4
B6. Combination of B1-B5	5	5	7	7	7	7	6	7	7	7	8
C. Tailored Tests											
C1. Combined contingent liabilities	5	5	5	5	5	5	5	5	5	5	5
C2. Natural disaster	5	5	6	6	5	6	6	6	6	5	6
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	15	15	15	15	15	15	15	15	15	15	15
Debt service-to-revenue ratio											
Baseline	7	6	6	6	6	5	5	5	5	5	4
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	7	6	6	5	4	3	3	2	1	0	-1
B. Bound Tests											
B1. Real GDP growth	7	7	6	6	6	6	5	6	5	5	5
B2. Primary balance	7	6	6	6	6	6	5	6	5	5	5
B3. Exports	7	7	7	8	7	7	7	7	7	7	6
B4. Other flows 3/	7	6	7	8	7	7	7	8	8	8	7
B5. Depreciation	7	8	8	7	6	6	6	6	5	5	5
B6. Combination of B1-B5	7	7	8	8	7	7	7	8	7	7	6
C. Tailored Tests											
C1. Combined contingent liabilities	7	6	6	6	6	6	6	6	5	5	5
C2. Natural disaster	7	6	7	6	6	6	6	6	6	6	5
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	18	18	18	18	18	18	18	18	18	18	18
Memorandum item:											
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	**0.0**

Sources: Country authorities; and staff estimates and projections.

1/ A bold value indicates a breach of the threshold.
2/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
3/ Includes official and private transfers and FDI.

Table 4. Myanmar: Sensitivity Analysis for Key Indicators of Public Debt 2019–29

(In percent)

	Projections 1/										
	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029
PV of Debt-to-GDP Ratio											
Baseline	35	35	36	37	38	39	41	42	41	41	40
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	35	35	36	36	38	39	40	42	42	42	42
B. Bound Tests											
B1. Real GDP growth	35	36	38	39	41	43	45	47	47	47	46
B2. Primary balance	35	36	38	39	40	42	43	44	43	43	42
B3. Exports	35	37	41	42	43	44	46	47	46	45	43
B4. Other flows 3/	35	39	43	44	45	46	47	48	47	46	45
B5. Depreciation	35	35	34	33	32	32	32	32	30	28	26
B6. Combination of B1-B5	35	35	35	35	36	37	39	40	40	39	38
C. Tailored Tests											
C1. Combined contingent liabilities	35	41	41	42	43	44	46	47	46	45	44
C2. Natural disaster	35	46	46	47	48	50	51	52	51	51	50
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
TOTAL public debt benchmark	55	55	55	55	55	55	55	55	55	55	55
PV of Debt-to-Revenue Ratio											
Baseline	197	195	197	198	201	205	209	212	208	201	196
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	197	196	197	198	199	202	206	210	208	204	204
B. Bound Tests											
B1. Real GDP growth	197	200	208	213	218	225	232	237	235	230	227
B2. Primary balance	197	202	211	212	213	217	220	223	218	211	205
B3. Exports	197	206	229	229	230	232	235	236	229	220	213
B4. Other flows 3/	197	217	239	238	238	241	243	243	236	226	219
B5. Depreciation	197	193	185	178	173	169	166	163	152	140	129
B6. Combination of B1-B5	197	192	195	189	192	196	200	203	198	192	187
C. Tailored Tests											
C1. Combined contingent liabilities	197	228	229	229	230	232	235	237	231	223	217
C2. Natural disaster	197	254	255	255	256	259	262	263	258	250	244
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Debt Service-to-Revenue Ratio											
Baseline	30	47	60	70	81	88	96	78	66	64	61
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	30	49	61	71	82	89	97	78	64	64	62
B. Bound Tests											
B1. Real GDP growth	30	48	62	74	86	95	104	86	74	74	71
B2. Primary balance	30	47	62	75	85	93	100	82	69	68	64
B3. Exports	30	47	60	72	82	90	97	79	68	66	63
B4. Other flows 3/	30	47	61	72	83	90	97	80	68	67	64
B5. Depreciation	30	45	56	64	74	81	88	71	59	58	54
B6. Combination of B1-B5	30	46	58	68	78	86	93	75	63	62	59
C. Tailored Tests											
C1. Combined contingent liabilities	30	47	73	81	92	100	105	86	73	72	68
C2. Natural disaster	30	48	83	90	102	111	115	96	82	80	77
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.

Sources: Country authorities; and staff estimates and projections.

1/ A bold value indicates a breach of the benchmark.

2/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.

3/ Includes official and private transfers and FDI.